

QC Report — Chloe's Biology Final Study Bundle

Run ID: 2026-05-29_biology-final Date assembled: 2026-05-16 Test: Friday, May 29, 2026 · 1–3 PM
· North Gym · Bertram / Deasy / Seatter Time-to-test at assembly: 13 days

Materials catalog — 28 files from Google Drive folder

All 28 files in the parent's Drive folder were inventoried. **Every meaningful file was read in full and incorporated.**

Lecture slides (teacher materials) — 8 files

#	File	Read?	Used for
1	Evolution 25-26.pptx.pdf	✓	Topic V content
2	Genetics.pptx (1).pdf	✓	Topic IV content
3	Ecology .pptx.pdf	✓	Topic VI content
4	Circulatory and Respiratory Systems.pptx.pdf	✓	Topic VII content
5	RNA and Protein Synthesis.pptx.pdf	✓	Topic II content
6	Evolution of Populations.pptx.pdf	✓	Topic V (Ch. 18)
7	Meiosis and Reproduction.pptx (1).pdf	✓	Topic III + VII repro
8	X Linked Inheritance CGPS.pptx (1).pdf	✓	Topic IV (X-linked)

Reading guides — 6 files

#	File	Read?	Student's answers filled in?
9	Ch. 12 Reading Guide.pdf	✓	NO (blank template)
10	Chapter 17 Reading Guide (1).pdf	✓	NO (blank template)
11	Ch. 12.4 Reading_ Meiosis.docx.pdf	✓	NO (blank template)
12	Chapter 18 Reading Guide.docx.pdf	✓	NO (blank template) — captured teacher exclusion
13	Ecology Reading Guide.pdf	✓	NO (blank template)
14	Chapter 14 RNA & Protein Synthesis Reading Guide.docx.pdf	✓	NO (blank template)

Worksheet KEYs + worksheets — 11 files

#	File	Read?
15	REVIEW GUIDE FOR BIOLOGY FINAL EXAM - Friday, May 29, 2026.pdf	✓ (scope document — drives entire bundle)
16	Human+Reproduction+Review+Sheet+KEY.pdf	✓
17	Transcription_Translation+Practice+KEY.pdf	✓ (OCR fuzzy — used to verify codon-wheel concepts)
18	Mutations+WS+Answer+Key.pdf	✓
19	Protein+Synthesis+Worksheet+KEY.pdf	✓
20	Reproductive+System+Worksheet+KEY.pdf	✓
21	Reproductive+System+Worksheet+.pdf	✓ (blank version, same content)
22	Dihybrid+Crosses+(1).pdf	✓
23	Circulatory & Respiratory System Modeling.docx.pdf	✓
24	Circulatory+and+Respiratory+System+-+Crossword+Labs.pdf	✓
25	PUNNETT SQUARE FRENZY.pdf	✓

Reference + misc — 3 files

#	File	Status
26	32+The+Circulatory+System-S.pdf (POGIL activity)	✓ Read
27	Codon+Wheel.pdf (amino acid abbreviations table)	✓ Read — standard 20 amino acids
28	Assignment submittedOpens a dialog...webloc	➡ Skipped — macOS browser bookmark, no content

Total content files: 27 of 28 ingested. 1 was a non-content browser shortcut.

Primary-source escalation status

Level	Status
Level 1: Summary only	✓ Skipped — we had the actual files
Level 2: Text extraction from primary PDFs	✓ Done for all 27 content files via Google Drive read_file_content MCP

Level	Status
Level 3: Image rendering of PDFs	✓ Done for 4 highest-priority slide decks (Circulatory & Respiratory, Meiosis & Reproduction, Genetics, Evolution of Populations). Image-only slides were rendered via PyMuPDF and visually inspected. Findings incorporated:
	Heart anatomy SVG was UPDATED to include the full set of labels the teacher marks "KNOW FOR FINAL" — aortic arch, pulmonary trunk, right vs left pulmonary arteries, right vs left pulmonary veins.
	New Watch-Out callout in the study guide listing all 13 heart-anatomy labels Chloe needs to identify on a diagram.
	New visual: 3 side-by-side selection curve diagrams (directional / stabilizing / disruptive). Previously only described in words.
	New visual: cladogram with worked example for how to read it.
	New quiz problem (Q29B) — the EXACT pedigree problem from the teacher's slide ("attached earlobes autosomal recessive — what is the genotype of individual II-3?") with worked answer.
	New glossary entries: epididymis, seminiferous tubules, seminal vesicle, aortic arch, pulmonary trunk, amniotic sac, amniotic fluid (all explicitly labeled on the teacher's slides).
	Confirmed: the blood-flow loop diagram on the teacher's slide ("KNOW FOR FINAL") matches my study guide's loop exactly.
	The other 4 slide decks (Evolution 25-26, Ecology, RNA, X-Linked) had text-extraction sufficient — visuals there were mostly stock textbook images already covered by my SVGs and content.
Level 4: Student-work audit	✗ NOT done — Chloe's filled-in homework was not provided. See below.

⚠ Major caveat: NO student-work audit was possible

The skill's #1 highest-leverage move is auditing the student's own completed work against the primary source — finding the wrong answers they've been studying with and surfacing them as "Watch Out!" corrections.

This was not possible for this run because Chloe's filled-in reading guides / homework / prior quizzes were not in the Drive folder. Every reading guide in the folder was a blank template (questions without student answers).

What that means for the bundle: - All deliverables use the CORRECT primary-source content (drawn from teacher slides + reading-guide questions + worksheet answer keys). - But there are no "Errors in My Notes" / "Watch Out for what you wrote on Q5" callouts personalized to Chloe's actual mistakes. - General "Watch Out" callouts based on classic teen-bio confusions (e.g., homologous vs. analogous, pulmonary artery oxygen content, incomplete dominance vs. codominance) are included — but they're not based on Chloe's specific errors.

What to do about it: If Chloe's actual completed work (reading guides, prior tests, homework) is available, send them and a follow-up run can produce the personalized audit. That second pass is where the biggest grade improvements usually come from.

Teacher-specified exclusion captured

From `Chapter 18 Reading Guide.docx.pdf`:

"SKIP EVOLUTION VERSUS GENETIC EQUILIBRIUM AND THE HARDY-WEINBERG PRINCIPLE"

This is honored in the bundle: - Hardy-Weinberg is NOT covered in the glossary, study guide, quiz bank, cheat sheet, or flashcards. - An explicit "DO NOT STUDY" callout flags it in the study guide, study plan, and Chloe's README.

Deliverables verified

Deliverable	Format	Status	Pages	Notes
Interactive study guide	Standalone HTML	✓	n/a (web)	All 6 SVGs inlined; no external dependencies; opens offline; mobile-responsive
Study guide PDF	PDF	✓	17	Under 25-page cap
Cheat sheet	PDF (2 pages)	✓	2	Exactly 2 pages ✓
Study plan	PDF + .docx	✓	3	13-day plan aligned to teacher's calendar
Glossary	PDF + .docx	✓	9	Every term from teacher's review guide
Quiz bank	PDF + .docx	✓	8	60+ questions across all 7 topics
Flashcards	.txt (Quizlet)	✓	n/a	125 cards tab-separated
Visuals	6 SVGs + 6 PDFs	✓	n/a	Central dogma, heart flow, respiratory path, meiosis phases, energy pyramid, Punnett squares
Consolidated PRINT PACK	Single PDF	✓	24	Cover + plan + study guide + cheat sheet. Under 25-page cap ✓
Per skill convention: glossary + quiz bank live as separate reference PDFs because including them would have pushed the pack over 25.				

Deliverable	Format	Status	Pages	Notes
QC report	Markdown + PDF (this)	✓	–	for_parent/
Grading scorecard	Markdown + PDF	✓	–	for_parent/

Format / distribution discipline checks

- ✓ No React / JSX / Tailwind / framework-dependent artifacts.
- ✓ HTML uses only vanilla CSS + native `<details>` element for quiz reveal. Opens in any browser, including iPhone Safari.
- ✓ All SVGs inlined into HTML — no external image dependencies.
- ✓ PDFs render correctly via WeasyPrint.
- ✓ .docx files generated for the markdown-source deliverables (parent can upload to Google Drive as Google Docs).
- ⚠ **Distribution NOT done from this sandbox:**
- GitHub Pages publication: this sandbox's GitHub MCP is restricted to `schein-ui/study-prep-skill` — a NEW repo for student-only viewing cannot be created from here. The standalone HTML and PDFs were committed to the development branch `claude/create-study-guide-vmLJv` instead. Parent can fork or download and host elsewhere.
- iCloud sync: not available — sandboxed Linux environment with no iCloud mount.
- Apple Mail / email draft: not available — no Mac AppleScript bridge.

Recommended distribution actions for the parent: 1. **Easiest:** Download the `student_downloads/` folder from the branch. AirDrop the `00_START_HERE.html` to Chloe's phone (or email it to her). Also email her `Chloe_Biology_Complete.pdf` for printing. 2. **For shared viewing:** Upload the contents of `student_downloads/` to a Google Drive folder and share with Chloe. 3. **For phone access:** Upload `00_START_HERE.html` to a personal GitHub Pages repo (it's a single self-contained file with no dependencies).

Known limitations / honest disclosure

1. **No student-work audit** (see above) — biggest gap.
2. **Image rendering of slide PDFs:** Done for the 4 highest-priority decks (Circulatory, Meiosis, Genetics, Evolution Pop). The remaining 4 decks (Evolution 25-26, Ecology, RNA, X-Linked) were not rendered — their visual content is mostly standard textbook figures already covered by my SVGs and content. If a test question hangs on a specific visual from one of those 4 unrendered decks, this guide may not flag it explicitly.
3. **No multi-expert agentloop review.** The bundle was produced by a single pass without the multi-expert grading loop. The grading scorecard (next file) reflects only an honest self-assessment, not multiple-expert sign-off.
4. **Polar bodies wording in oogenesis.** Most 9th-grade textbooks describe oogenesis as producing 1 ovum + 3 polar bodies (one in Meiosis I → polar body + secondary oocyte, then secondary oocyte → 1 ovum + 1 polar body in Meiosis II = 1 ovum + 2 polar bodies, OR + 3 if the first polar body also divides).

The bundle uses the common simplified "1 egg + 3 polar bodies" framing matching the Meiosis & Reproduction slides Chloe was taught.

Files to send to Chloe vs. files for parent

Send to Chloe (entire `student_downloads/` folder): - README_for_Chloe.txt - 00_START_HERE.html - All `Chloe_*.pdf` and `Chloe_*.docx` - `Chloe_Flashcards.txt` - `diagrams_pdf/` folder

Keep in `for_parent/` (do NOT send to Chloe — she should see study material, not the audit of that material): - QC_Report.pdf (this document) - Grading_Scorecard.pdf